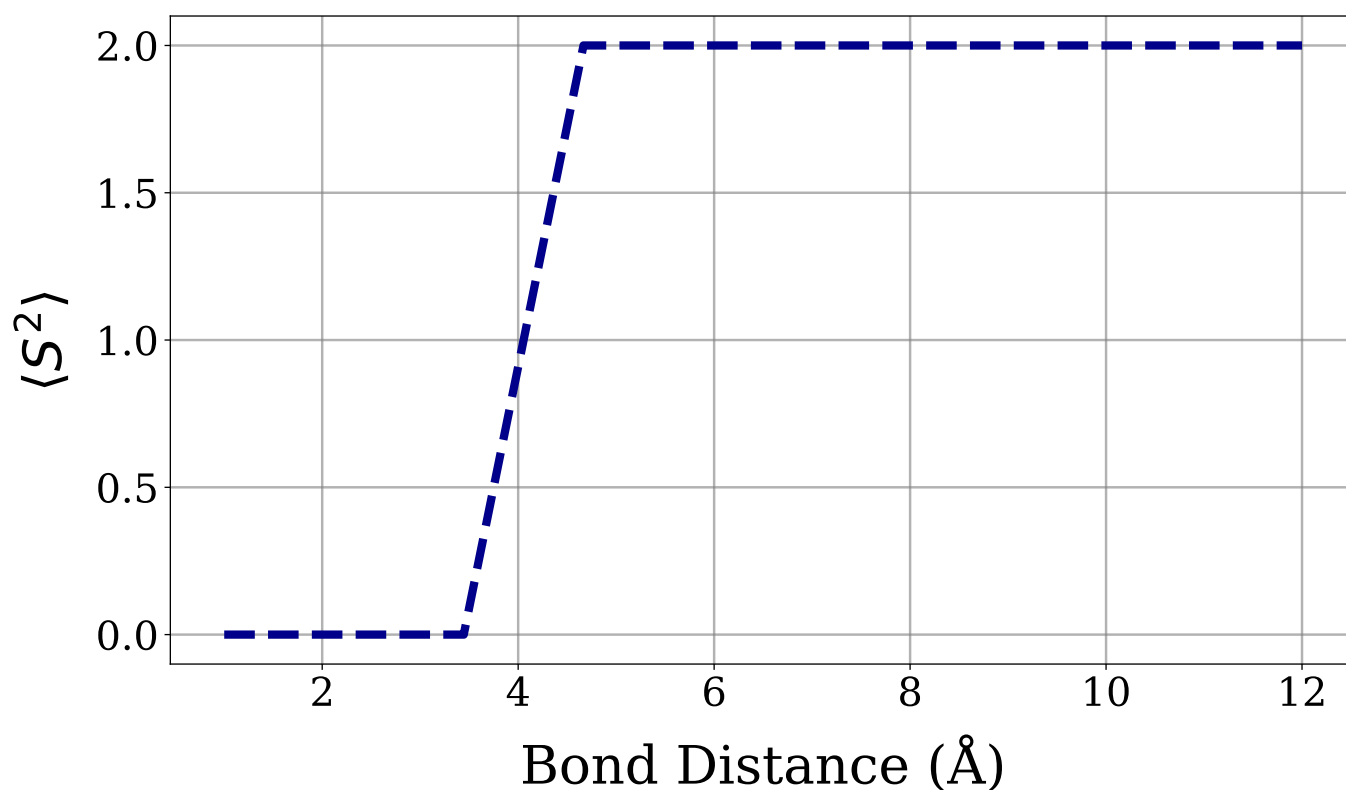


Plot Analysis

1) Effect of Regularization

a) Identifying Coulson-Fischer Point for dihydrogen

Identification of the Coulson-Fischer Point H₂ (cc-pVTZ,GHF)



Question

At which bond distance the spin symmetry breaks?

Hypothesis

Few angstroms away from the equilibrium we will see $\langle S^2 \rangle$ becomes non zero.

Details

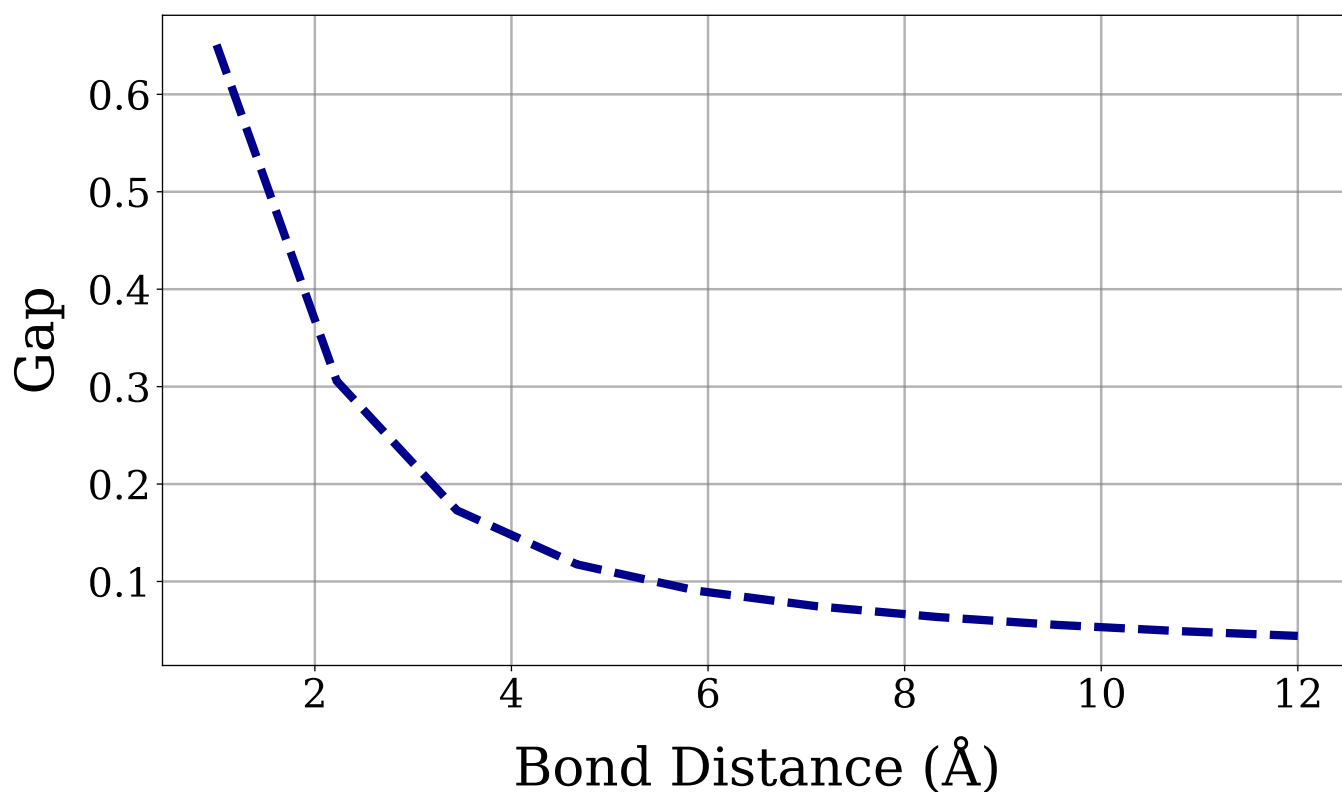
- System: H2
- Initialization: GHF
- Units: Angstroms

Obeservation

Spin symmetry breaks after stretching the bond beyond ~ 5 Angstroms.

b) Effect of bond stretching to the HOMO-LUMO gap for dihydrogen

HOMO-LUMO Gap vs Bond Stretching H₂ (cc-pVTZ)



Question

Does near CF point the HOMO-LUMO gap becomes smaller and become worse afterwards?

Hypothesis

- Near CF point the gap should be small and further away from the point it should become even smaller.

Details

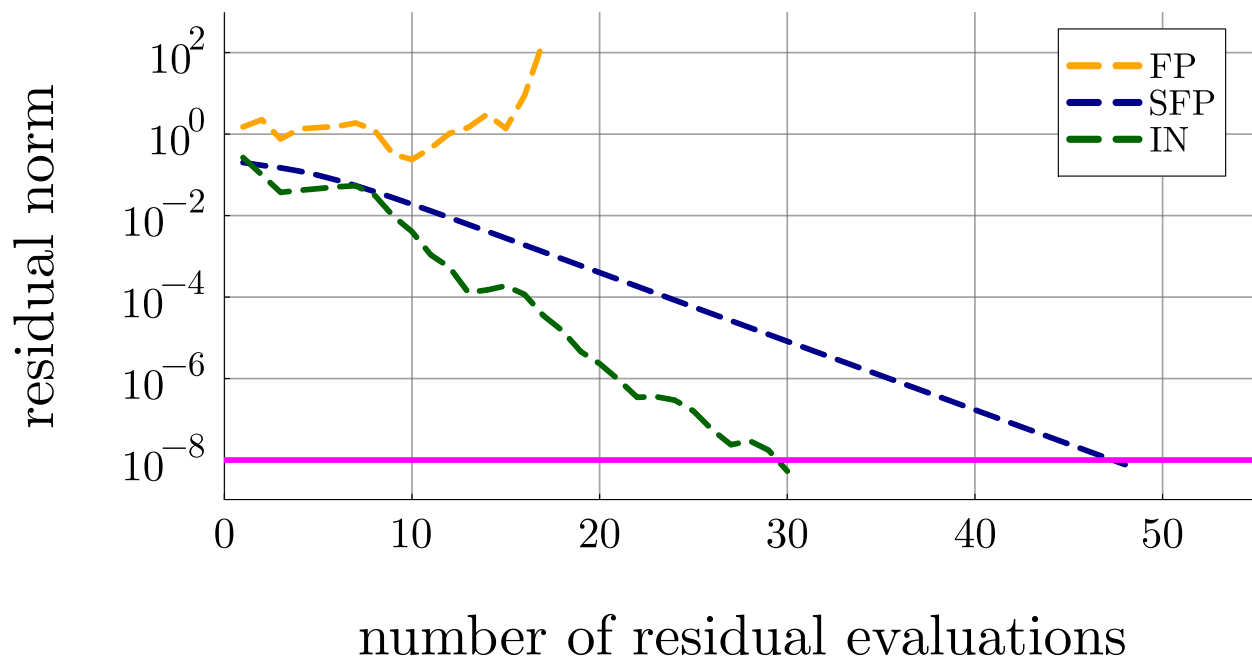
- System: H₂
- Initialization: RHF
- Units: Angstroms

Obeservation

- Near CF point the gap is ~ 0.1.
- Away from CF point it becomes smaller.
- Around 20 angstroms the gap bemoes larger (I don't know why)

c) Effect of Regularization

Effect of Regularization dihydrogen (7Å,cc-pVTZ)



Question

How effective is regularization (level shift) for the convergence and convergence rate for a small-gap system, and how does it compare to INK?

Hypothesis

- FP should diverge.
- SFP and INK should converge.
- INK should converge at a faster rate.

Details

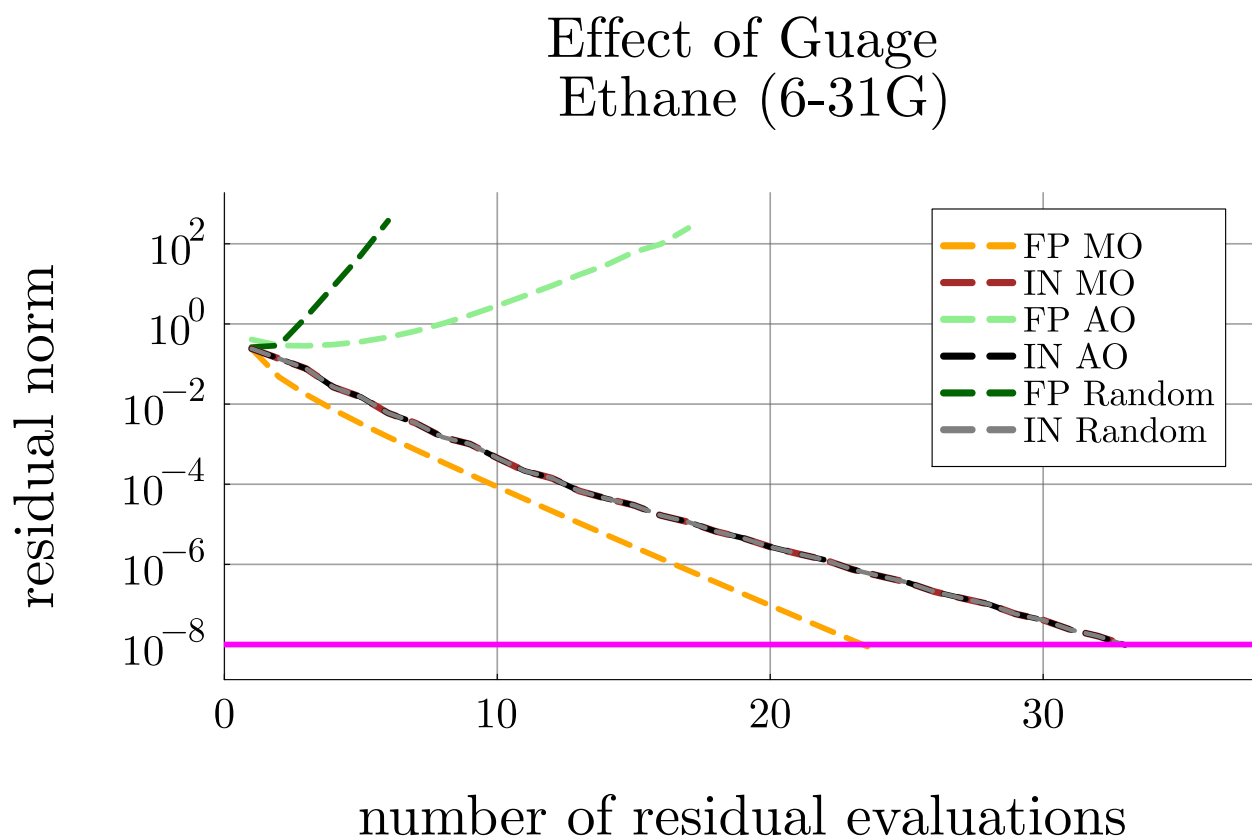
- System: H₂-CF at 7 Angstrom, cc-pVTZ (small-gap case)
- Gauge: MO

Observation

- FP suffers for few iterations with having amplitude difference ~ 1 in 2 norm and then diverges.
- SFP took 46 iterations to converge.
- INK showed few fluctuations at start and then took 31 iterations to converge.
- For a system: H₂-CF at 10 Angstrom, cc-pVTZ (evensmaller-gap case) any of the solvers does behave as usual.
- PNK, SFP, SFP+DISS converge to different solutions.

- PNK converge to different solutions with different rates when applied to different gauges.

2) Effect of Gauge



Question

How sensitive are FP and INK to gauge choice (MO, AO, random) for a large-gap system?

Hypothesis

- FP should converge in MO and diverge in AO and random (assuming bigger rotation from MO).
- INK should converge in all gauges with same numbers.

Details

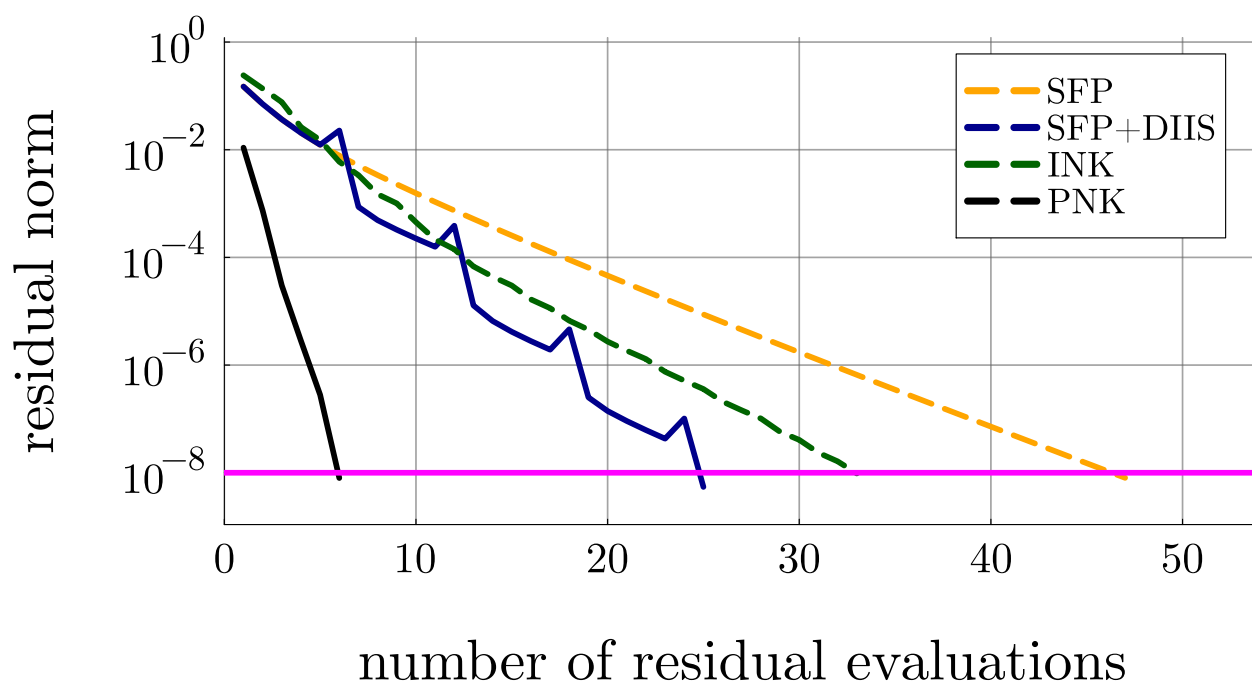
- System: Ethane, 6-31G (large-gap case)
- Gauges compared: MO, AO, random gauge

Observation

- FP converges only in MO.
- INK converges in all gauges with same numbers.
- FP converge faster compared to INK in MO basis. (My guess is the samml shift I apply to FP to avoid zero divisions would do the this)

3) Effect of Jacobian Inverse

Effect of Jacobian Inverse Ethane (6-31G)



Question

What is effect on the convergence and the rate based on how exact are we constructing the Jacobian and how we construct it?

Hypothesis

- All solvers should converge.
- SFP should be slower than INK.

Details

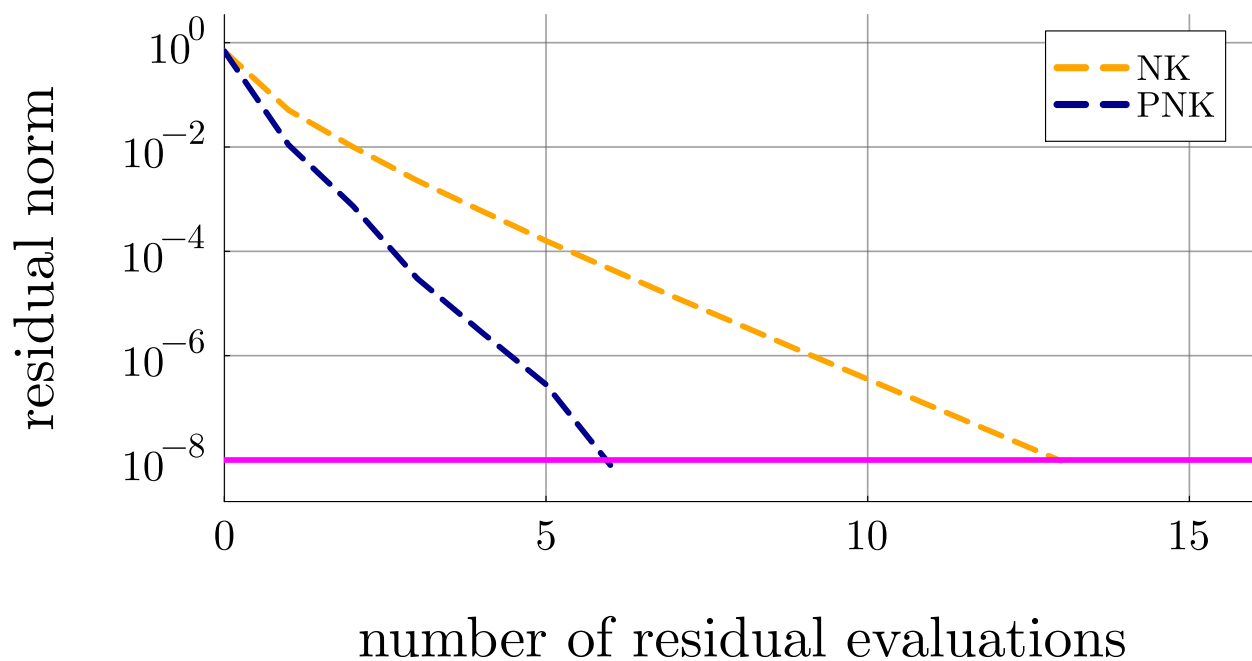
- System: Ethane, 6-31G (large-gap case)
- Gauges compared: MO

Observation

- All solvers should converge.
- PNK was the fastest.
- SFP was the slowest.
- INK was faster than SFP but with DIIS SFP is faster than INK.

5) Effect of Preconditioning

Effect of Preconditioning Ethane (6-31G)



Question

What is the effect on NK by preconditioning in an arbitrary gauge?

Hypothesis

PNK should be faster than NK.

Details

- System: Ethane, 6-31G (large-gap case)
- Gauges compared: Random

Observation

- PNK is significantly faster.